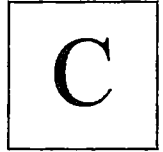


--	--	--	--	--	--	--	--



B.Tech. Degree V Semester Regular Examination November 2021

ME 19-205-0503 MACHINING SCIENCE AND MACHINE TOOLS (2019 Scheme)

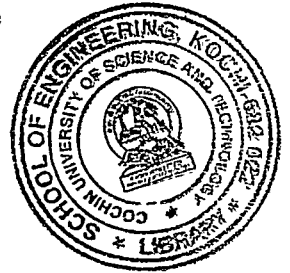
Time: 3 Hours

Maximum Marks: 60

PART A (Answer *ALL* questions)

(8 × 3 = 24)

- I. (a) What are the conditions that would allow a continuous chip to be formed in metal cutting?
- (b) Write a short note on thermal aspects in machining.
- (c) What are the various elements present in a machine tool for the purpose of generating any specified surface?
- (d) How do you classify various manufacturing machines based on the motions of the machine axes?
- (e) Explain one method used for taper turning in a lathe.
- (f) Give a neat sketch of a slotter and mark its important parts.
- (g) What are the advantages and limitations of using centerless grinding?
- (h) Explain the applications of upmilling and down milling.



PART B

(4 × 12 = 48)

- II. (a) From the machining performance view point which type of chip is preferred? Explain with suitable justification. (4)
- (b) Derive an expression for the main cutting force in orthogonal cutting in terms of work material properties and cutting process parameters. (3)
- (c) Explain the geometry of a twist drill with neat sketches. (5)
- OR**
- III. (a) Explain the differences between orthogonal and oblique cutting. (3)
- (b) How do you define tool life? Explain the parameters that control the tool life of a single point cutting tool. (5)
- (c) What are the main regions of heat generation in metal cutting? Explain with the help of suitable sketches. (4)
- IV. (a) How do you select a machine tool for a given application? Give your answer with an application. (6)
- (b) Explain the classification of machine tools based on production capacity. Give the relative advantages and applications. (6)
- OR**
- V. (a) What is the function served by a lead screw in a machine tool? Explain with the help of suitable figures. (6)
- (b) Explain linear motion elements used in machine tools. Give sketches. (6)

(P.T.O.)

BTS-V(R)-11.21-0949

- VI. (a) How is a lathe specified? Explain with a neat sketch the relevance of each of the specification points. (6)
- (b) A taper pin of length 80 mm has a taper length of 48 mm. The longer diameter of taper is 83 mm and the smaller diameter is 73 mm. Determine: (6)
- Taper in mm/metre and in degrees.
 - The angle to which the compound rest should be set up.
 - The tail stock setting over.

OR

- VII. (a) Give similarities and differences between shaping and planing machines with respect to constructional features, applications and working. (4)
- (b) Sketch an example component suited for Capstan lathe with dimensions. (3)
- (c) Calculate the machining time required for shaping a 250 mm long and 200 mm wide flat surface on a shaper using a cutting speed of 40 m/min and a feed 0.5 mm/stroke. The depth of cut is 4 mm. Calculate the material removal rate and the power required. (5)
- VIII. (a) Describe the differences between a lathe and the milling machine in terms of the types of surfaces generated and the types of tools used. (3)
- (b) What are the differences between compound indexing and differential indexing? Explain the relative merits. (4)
- (c) A C50 steel flat surface of 100 mm × 250 mm is to be machined on a horizontal axis milling machine. An HSS slab mill of 100mm diameter and 150 mm width is to be used for the purpose. The milling cutter has 8 teeth. Calculate the machining time assuming that entire stock can be removed in one depth of 2 mm. (5)

OR

- IX. (a) How is grinding different from other machining operations? Explain its applications in view of its capabilities. (4)
- (b) How is the abrasive selected for a grinding operation? Indicate the reasons for selection. (3)
- (c) Give any four points to be kept in mind to decide the clamping system in jigs and fixtures. (5)
